Machine Learning: k-Means and PCA for Regime Detection

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1. Abstract

In this project, we apply k-means clustering and PCA to solve the problem of market regime detection in multi-dimensional financial time-series. We compare these techniques against other regime detection methods: the Hidden Markov Model with Gaussian Mixtures (GMMHMM), sliced-Wasserstein k-means clustering, and a simple peak-to-trough-based regime labelling method. We evaluate the quality of each detection method using a trading strategy that goes long when the label indicates a bullish regime and flat otherwise, comparing cumulative returns across methods and regions (America, Europe, Asia).

2. Introduction

Financial markets, especially the stock market, have complexities. Due to various factors, such as macroeconomic recession, debt crisis, and policy changes, the market will show different regimes, such as bull and bear markets. Changes in market conditions often lead to fluctuations in stock prices. In different market regimes, investors need to use different trading strategies to increase returns and reduce losses. Therefore, detecting market regimes accurately plays an important role in making investment decisions and controlling trading risks.

K-means clustering and PCA are combined to identify market patterns in multidimensional financial time series. The trading strategy used to go long when the label shows bull and keep the position closed otherwise. Cumulative return, Annualized return, Annualized volatility and Sharpe ratio are used to measure the performance of trading strategies based on the results of market regime detection. The performance of the PCA k-means, the Hidden Markov Model with Gaussian Mixtures (GMMHMM), sliced-Wasserstein k-means clustering, and a simple peak-to-trough-based regime labelling method and long only were compared.

3. Data Processing and Feature Engineering

We track 9 major stock indices across 3 regions to develop our analysis.

Table 1: Index universe by region.

Region	Index	Ticker
America	S&P 500 / Dow Jones / Nasdaq	`^GSPC`, `^DJI`, `^IXIC`
Europe	FTSE 100 / CAC 40 / DAX	`^FTSE`, `^FCHI`, `^GDAXI`
Asia	KOSPI / Nikkei 225 / Hang Seng	`^KS11`, `^N225`, `^HSI`

3.1. Raw Data Structure

Each index is downloaded from Yahoo Finance with daily OHLCV columns. After merging, the full DataFrame uses a MultiIndex on columns of the form `(index_name, OHLCV_field)`.

3.2. Feature Engineering

From the closing prices we compute the following features for each index:

Log-returns. Over multiple horizons (1, 5, and 21 trading days):

$$r_{t,p} = \ln\left(\frac{P_t}{P_{t-p}}\right). \tag{1}$$

Rolling annualised volatility. With windows of 21 and 63 days:

$$\sigma_{t,w} = \mathrm{std}(r_{t-w:t}) \times \sqrt{252}\,. \tag{2}$$

Rolling correlation. For each pair of indices X and Y at time t with window w:

$$\rho_{X,Y,t} = \frac{\sum_{i=0}^{w-1}(r_{X,t-i} - \bar{r}_X)(r_{Y,t-i} - \bar{r}_Y)}{\sqrt{\sum_{i=0}^{w-1}(r_{X,t-i} - \bar{r}_X)^2 \ \sum_{i=0}^{w-1}(r_{Y,t-i} - \bar{r}_Y)^2}} \tag{3}$$

Momentum Z-score. Measures mean-reversion by normalising distance to the moving average:

$$Z_{t,w} = \frac{P_t - \mathrm{MA}_{t,w}}{\sigma_{t,w}} \tag{4}$$

RSI. The Relative Strength Index captures overbought and oversold conditions:

$$\mathrm{RSI}_t = 100 - \frac{100}{1 + \frac{\text{Average Gain}_t}{\text{Average Loss}_t}} \tag{5}$$

where Average Gain_t and Average Loss_t are computed over 14 days.

Building the feature set. We merge all features (log-returns, rolling volatility, correlation, z-score, RSI) into a single DataFrame. Per index: 3 log-returns + 2 volatilities + 1 z-score + 1 RSI = 7 features; across 3 indices this gives 21 features, plus 3 pairwise correlations, yielding a raw feature matrix of 24 columns. The number of rows equals the trading days minus a 63-day warm-up period. We standardise features and apply PCA retaining 95% of variance.

Table 2: Feature set dimensions after PCA (95% variance retained).

Region	Raw dimensions	After PCA
America	$(3460, 24)$	$(3460, 7)$
Europe	$(3523, 24)$	$(3523, 10)$
Asia	$(3561, 24)$	$(3561, 14)$

3.3. Pipeline Summary

Yahoo Finance API → Raw OHLCV → Merged DataFrame → Close prices

↓

Feature Engineering (returns, volatility, correlation, z-score, RSI)

↓

Standardisation → **PCA** → **K-Means Clustering**

Figure 1: Data pipeline from raw download to regime labels.

4. Mathematical Foundations

4.1. K-Means Clustering

K-means partitions data into K clusters by minimising the within-cluster sum of squares (WCSS):

$$\text{WCSS} = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2 \tag{6}$$

where C_k is the set of points in cluster k and μ_k is its centroid. In high-dimensional settings, the curse of dimensionality can limit performance; we therefore reduce dimensionality via PCA before clustering.

4.2. Principal Component Analysis (PCA)

PCA finds directions of maximum variance via the singular value decomposition of the centred data matrix:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \tag{7}$$

where $\mathbf{U}$ contains left singular vectors, $\mathbf{\Sigma}$ is a diagonal matrix of singular values, and $\mathbf{V}$ contains the principal components. We retain the top d^* components capturing $\geq 95\%$ of variance.

4.3. Alternative Regime Detection Methods

To provide a broader comparison, we also evaluate several benchmark regime-detection methods with different modelling assumptions.

4.3.1 Gaussian-Mixture Hidden Markov Model (GMM-HMM)

The Hidden Markov Model assumes that the market evolves through a sequence of unobserved states, with state transitions governed by a first-order Markov chain:

$$P(z_t = j \mid z_{t-1} = i) = A_{ij} \tag{8}$$

where A_{ij} denotes the probability of moving from hidden state i to hidden state j. Conditional on the hidden state, the observed feature vector x_t is modelled as a Gaussian mixture:

$$p(x_t \mid z_t = k) = \sum_{m=1}^{M} \pi_{k,m} \mathcal{N}(x_t \mid \mu_{k,m}, \Sigma_{k,m}) \tag{9}$$

where $\pi_{k,m}$ are mixture weights, and $\mu_{k,m}, \Sigma_{k,m}$ are the corresponding mean vectors and covariance matrices. This allows the model to capture both temporal persistence in market regimes and more flexible within-state feature distributions.

4.3.2 Sliced-Wasserstein K-Means

Unlike standard K-means, sliced-Wasserstein K-means compares rolling return windows through their empirical distributions rather than Euclidean distances between daily feature vectors. Given two distributions μ and ν, the sliced-Wasserstein distance is defined as

$$SW_p(\mu, \nu) = \left(\int_{\mathbb{S}^{d-1}} W_p^p(P_\theta \# \mu,\ P_\theta \# \nu)\, d\sigma(\theta) \right)^{1/p} \tag{10}$$

where P_θ denotes projection onto direction θ, $P_\theta \# \mu$ is the projected distribution, and W_p is the one-dimensional Wasserstein distance. In practice, this is approximated using a finite number of

random projections. This method is therefore more sensitive to changes in distributional shape and tail-risk structure across rolling windows.

4.3.3 Peak-to-Trough and Long-Only Benchmarks

As a simple rule-based benchmark, peak-to-trough labelling identifies crisis periods using drawdowns from the running peak of the cumulative return series:

$$DD_t = \frac{C_t - \max_{s \le t} C_s}{\max_{s \le t} C_s} \tag{11}$$

where C_t denotes the cumulative portfolio value at time t. A period is labelled as a crisis regime whenever DD_t falls below a chosen threshold. Finally, a long-only strategy is included as a passive benchmark that remains fully invested throughout the sample period.

5. Proposed Algorithm

Algorithm 1 K-Means Regime Detection Pipeline

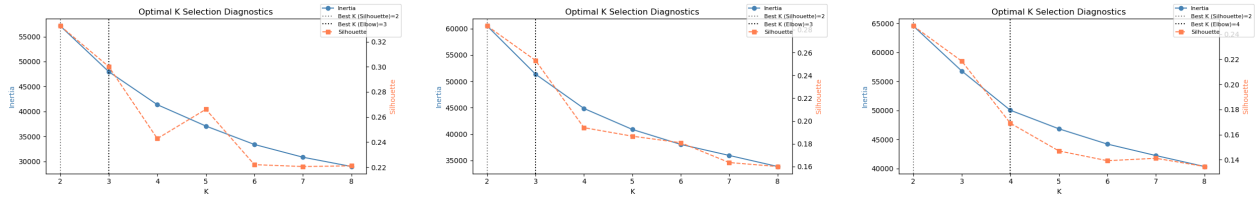
Require: Closing prices $\mathbf{P} \in \mathbb{R}^{T \times N}$, variance threshold α, range $[K_{\min}, K_{\max}]$
Ensure: Regime labels $\ell_t \in \{0, \ldots, K^* - 1\}$, ordered by mean return

— **Stage 1: Feature Engineering** —
for each index $n = 1, \ldots, N$ **do**
  Compute log-returns $r_{t,p}^{(n)}$ for $p \in \{1, 5, 21\}$
  Compute rolling volatility $\sigma_{t,w}^{(n)}$ for $w \in \{21, 63\}$
  Compute momentum z-score $Z_t^{(n)}$ and $\mathrm{RSI}_t^{(n)}$
end for
for each pair (i, j) with $i < j$ **do**
  Compute $\rho_t^{(i,j)} \leftarrow \mathrm{RollingCorr}_{63}(r^{(i)}, r^{(j)})$
end for
Concatenate into $\mathbf{F} \in \mathbb{R}^{T' \times D}$; drop warm-up rows
— **Stage 2: Standardisation & PCA** —
Z-score normalise each feature; compute SVD of $\tilde{\mathbf{F}}$
Select $d^* = \min\{d : \sum_{i=1}^{d} \sigma_i^2 / \sum_{i=1}^{D} \sigma_i^2 \ge \alpha\}$; project $\mathbf{X} \leftarrow \tilde{\mathbf{F}}\, \mathbf{V}_{:,1:d^*}$
— **Stage 3: Optimal K & Clustering** —
for $K = K_{\min}, \ldots, K_{\max}$ **do**
  Run K-Means; record inertia $\mathcal{I}(K)$ and silhouette $\mathcal{S}(K)$
end for
$K^* \leftarrow \arg\max_K \mathcal{S}(K)$; run final K-Means with K^*
— **Stage 4: Regime Ordering** —
Compute mean basket return $\bar{r}_k$ per cluster; relabel 0 = bearish, $K^* - 1$ = bullish
return $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_{T'})$

6. Results

6.1. Optimal K Selection

We applied k-means for $K \in \{2, \ldots, 10\}$ on the PCA-reduced feature space and computed inertia and silhouette scores. As shown in Figure 2, the silhouette score is maximised at $K = 2$ across all regions, suggesting two dominant regimes over 2006–2020.This result suggests that, given the sample period and feature design adopted in this study, market states are primarily characterised by two dominant structures: a relatively weak bearish regime and a relatively strong bullish regime. Relative to larger values of K, the two-regime specification achieves better cluster separation while avoiding the introduction of excessive noise and potential overfitting. It also preserves the stability and interpretability of the detected regimes.

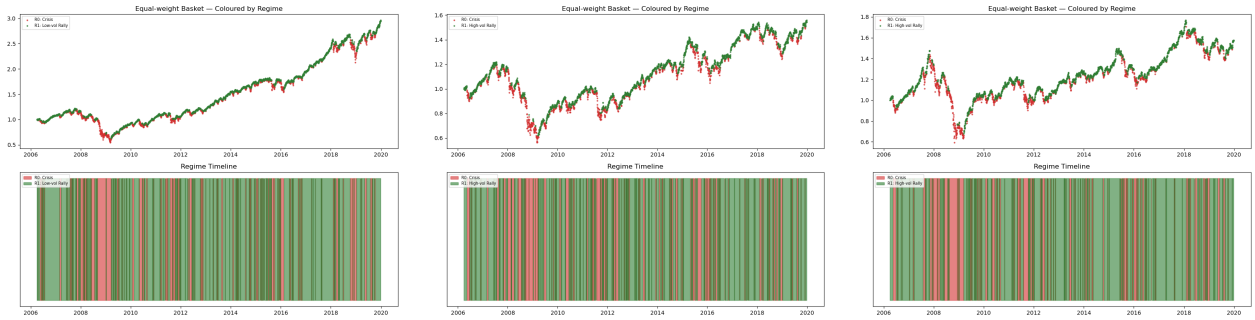


(a) American indices. **(b)** European indices. **(c)** Asian indices.

Figure 2: Inertia and silhouette scores for different K values across regional indices.

6.2. Regime Detection Results

Figure 3 shows the detected regimes overlaid on an equally weighted basket of indices for each region. In the figure, the red segments represent the crisis regime identified by the model, while the green segments indicate more favourable regimes. These periods broadly align with major historical episodes of market stress, including the 2008 global financial crisis, the 2011 European debt crisis, and the 2015–2016 period of heightened market volatility.

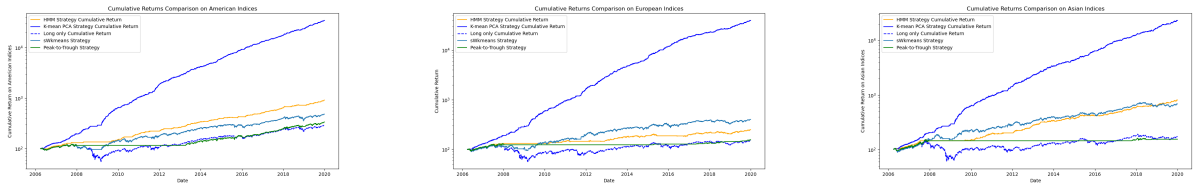


(a) American indices. **(b)** European indices. **(c)** Asian indices.

Figure 3: Regime detection on equally weighted baskets for each region.

6.3. Trading Strategy Comparison

To evaluate regime quality, we implement a long/flat strategy: go long in bullish regimes, hold cash otherwise. We benchmark PCA K-Means against GMMHMM, sliced-Wasserstein k-means, peak-to-trough labelling (20% drawdown threshold), and a long-only baseline. All strategies start with $100 over March 2006 – January 2020.



(a) American indices. **(b)** European indices. **(c)** Asian indices.

Figure 4: Cumulative returns of trading strategies based on different regime detection methods.

The PCA K-Means strategy significantly outperforms all alternatives across every region, as summarised in Table 3.

Table 3: Backtest performance summary across regional indices.

Region	Metric	PCA K-Means	HMM	sWK-Means	Peak-to-Trough	Long Only
American	Cumulative Return (%)	34,808.39	915.48	483.31	337.18	293.05
	Annualised Return (%)	53	16	12	9	8
	Annualised Volatility (%)	10	6	15	10	19
	Sharpe Ratio	4.37	2.74	0.85	0.95	0.51
European	Cumulative Return (%)	39,532.73	247.17	401.41	155.08	149.26
	Annualised Return (%)	53	7	10	3	3
	Annualised Volatility (%)	11	7	15	6	20
	Sharpe Ratio	3.91	0.94	0.72	0.57	0.24
Asian	Cumulative Return (%)	23,225.90	810.79	687.00	155.36	173.54
	Annualised Return (%)	47	15	15	3	4
	Annualised Volatility (%)	10	8	14	6	19
	Sharpe Ratio	3.76	1.90	1.04	0.54	0.30

Looking at the performance metrics, the PCA K-Means strategy achieves annualised returns of 47–53% with Sharpe ratios above 3.7 across all regions, while alternatives lag significantly behind. The long-only benchmark delivers much lower returns and Sharpe ratios. We can therefore see that PCA K-means capture market regimes in a highly granuar way, as opposed to other methods which are either too noisy (sW k-means) or too coarse (peak-to-trough). The HMM performs decently but still falls short of the PCA K-Means approach, likely due to its parametric assumptions and sensitivity to regime duration. However, while we present performance metrics for trading strategies, those results have to be taken as a descriptive accuracy rather than a predictive accuracy, as for all regime detection methods, we have high look-ahead bias when labelling regimes. This is because we are using the entire dataset to detect regimes, and then backtesting a strategy on the same dataset. Therefore, while the PCA K-Means method shows superior performance in this backtest, further work would be needed to evaluate its predictive power in an out-of-sample context.

Method-level differences. From a modelling perspective, these performance differences are consistent with the mechanisms of the methods. PCA K-Means clusters daily observations in a denoised, PCA-reduced feature space built from returns, volatility, correlations, momentum z-scores, and RSI, and the resulting regimes are then ordered by mean return. This makes the method particularly effective at separating market states that are directly relevant for trading. By contrast, GMM-HMM uses the same feature set but imposes a latent-state structure with probabilistic transitions, allowing it to capture regime persistence over time at the cost of stronger modelling assumptions. Sliced-Wasserstein K-Means instead clusters rolling return windows based on distributional shape and tail-risk structure, which makes it more sensitive to local changes but also more prone to noisy regime switching. Peak-to-trough labelling is more interpretable but relies only on drawdown information, while the long-only strategy provides a passive benchmark with no regime adaptation.

Regional differences. The regional results in Table 3 also highlight meaningful differences across markets. In the American indices, the advantage of PCA K-Means is likely supported by the strong co-movement among the S&P 500, Dow Jones, and Nasdaq, which allows PCA to extract common market structure effectively; HMM also performs relatively well here, suggesting clearer regime persistence. In the European indices, PCA K-Means achieves the largest relative advantage over the benchmarks, which is consistent with the more pronounced structural stress episodes in the sample, particularly the 2011 European debt crisis. In the Asian indices, although PCA K-Means remains the strongest method, the gap relative to HMM and sWK-Means is narrower. This is consistent with the higher retained PCA dimensionality reported earlier, suggesting a more heterogeneous market structure across the KOSPI, Nikkei 225, and Hang Seng, where alternative methods can still capture

part of the regime variation.

6.4. Hyperparameter tuning

To further validate the PCA K-Means specification, we performed a grid search over several key hyperparameters for each regional dataset. Specifically, we varied the number of regimes, whether PCA was applied, the retained PCA variance, and alternative return and volatility horizons. The hyperparameter grid is shown in Table 4.

Table 4: Hyperparameter grid for PCA K-Means tuning.

Hyperparameter	Values
n_regimes	$\{2, 3, 4, 5\}$
`use_pca`	{`True`, `False`}
`pca_variance`	$\{0.90, 0.95\}$
`return_periods`	{[1,5,21], [1,5,21,63]}
`vol_windows`	{[21,63], [10,21,63]}

Each specification was evaluated using both the silhouette score and the Sharpe ratio of the corresponding regime-based trading strategy. The silhouette score measures how clearly separated the detected clusters are, while the Sharpe ratio measures the risk-adjusted return of the resulting strategy. To balance clustering quality and trading usefulness, we defined a combined score as an equally weighted average of the silhouette score and a rescaled Sharpe ratio:

$$\text{Combined Score} = 0.5 \times \text{Silhouette} + 0.5 \times \frac{\text{Sharpe}}{3}. \tag{12}$$

Table 5: Best hyperparameter configuration by region.

Region	K	PCA	PCA Var.	Return Periods	Vol Windows	Silhouette	Sharpe	Combined
America	2	True	0.90	[1,5,21]	[21,63]	0.3227	4.5167	0.9141
Europe	2	True	0.90	[1,5,21]	[21,63]	0.2754	4.1912	0.8362
Asia	2	True	0.90	[1,5,21]	[21,63]	0.2029	4.3604	0.8282

The tuning results are remarkably consistent across regions. The detailed tuning results for each region are reported in the Appendix. As shown in Table 5, the best-performing specification is the same in all three markets: $K = 2$, with PCA enabled, 90% retained variance, return periods of $[1, 5, 21]$, and volatility windows of $[21, 63]$. This strongly supports the main model specification and confirms that a simple two-regime structure is most effective under the current feature design. The consistency of the top-ranked configuration also suggests that PCA provides a useful, though moderate, improvement. In all three markets, the best non-PCA specification ranks just below the top PCA-based models, indicating that the raw feature set is already informative, while PCA mainly helps by removing redundant variation and producing a more compact representation. Similarly, retaining 90% of variance slightly outperforms retaining 95%, which suggests that a more aggressive dimensionality reduction can reduce noise without losing the key structure needed for clustering.

The tuning results also highlight meaningful regional differences. The American market achieves the highest silhouette and combined score, indicating the clearest regime separation and the strongest overall clustering structure. Europe ranks second, while Asia has the lowest silhouette score, although its Sharpe ratio remains strong. This pattern is consistent with the earlier interpretation that the Asian indices are more heterogeneous, making their regimes harder to separate cleanly in feature space, while the stronger co-movement in the American indices leads to more clearly defined market states.

Adding a 63-day return horizon or a 10-day volatility window does not improve the best result. This suggests that the baseline feature set already captures the main regime information, while additional short- or long-horizon features are more likely to introduce complexity than to improve performance.

7. Conclusion

We have presented a PCA-enhanced k-means clustering pipeline for market regime detection across American, European, and Asian equity indices. The method consistently identifies two dominant regimes (bullish and bearish) that align well with known periods of market stress. When used to drive a simple long/flat trading strategy, the PCA K-Means approach delivers substantially higher cumulative returns and Sharpe ratios than alternative methods (GMMHMM, sliced-Wasserstein k-means, peak-to-trough labelling) and the long-only benchmark across all three regions. These results suggest that combining dimensionality reduction with unsupervised clustering provides a robust framework for capturing market dynamics. Additionally, hyperparameter tuning results show that most of the model specifications are optimal in all three markets, with PCA mainly providing useful improvements by removing redundant variation. However, more aggressive dimensionality reduction can further reduce noise and preserve key structures in high-dimensional time series. The results also show that the market regimes shows obvious regional differences, and the current feature set can well reflect the market regimes. One limitation of this study is that there is no limit on the number of trades, and regimes can change very quickly when the market is volatile, and frequent trading incurs a lot of costs, which can affect the performance of the trading strategy. Future work could explore out-of-sample regime prediction, rolling re-estimation windows, and the inclusion of additional asset classes to further validate the approach.

A. Detailed Hyperparameter Tuning Results

This appendix reports the top 10 hyperparameter configurations for each regional dataset, ranked by combined score.

A.1. American Market

Table 6: Top 10 hyperparameter configurations for the American dataset.

Rank	K	PCA	PCA Var.	Return Periods	Vol Windows	Silhouette	Sharpe	Combined
1	2	True	0.90	[1,5,21]	[21,63]	0.3227	4.5167	0.9141
2	2	True	0.95	[1,5,21]	[21,63]	0.3091	4.5008	0.9047
3	2	False	N/A	[1,5,21]	[21,63]	0.2947	4.5153	0.8999
4	2	True	0.90	[1,5,21]	[10,21,63]	0.3314	4.1693	0.8606
5	2	True	0.95	[1,5,21]	[10,21,63]	0.3148	4.1617	0.8510
6	2	True	0.90	[1,5,21,63]	[21,63]	0.3286	4.0854	0.8452
7	2	False	N/A	[1,5,21]	[10,21,63]	0.3030	4.1617	0.8451
8	2	True	0.95	[1,5,21,63]	[21,63]	0.3155	4.0989	0.8409
9	2	False	N/A	[1,5,21,63]	[21,63]	0.3018	4.0989	0.8341
10	3	True	0.90	[1,5,21]	[21,63]	0.3191	3.6943	0.7753

A.2. European Market

Table 7: Top 10 hyperparameter configurations for the European dataset.

Rank	K	PCA	PCA Var.	Return Periods	Vol Windows	Silhouette	Sharpe	Combined
1	2	True	0.90	[1,5,21]	[21,63]	0.2754	4.1912	0.8362
2	2	True	0.95	[1,5,21]	[21,63]	0.2618	4.1837	0.8282
3	2	False	N/A	[1,5,21]	[21,63]	0.2508	4.1650	0.8196
4	3	True	0.90	[1,5,21,63]	[10,21,63]	0.2498	3.8258	0.7625
5	3	True	0.95	[1,5,21,63]	[10,21,63]	0.2319	3.7807	0.7461
6	2	True	0.90	[1,5,21,63]	[21,63]	0.2818	3.6083	0.7423
7	3	False	N/A	[1,5,21,63]	[10,21,63]	0.2179	3.7891	0.7405
8	2	True	0.95	[1,5,21,63]	[21,63]	0.2691	3.6115	0.7365
9	2	False	N/A	[1,5,21,63]	[21,63]	0.2562	3.6381	0.7344
10	4	True	0.90	[1,5,21]	[10,21,63]	0.2040	3.7728	0.7308

A.3. Asian Market

Table 8: Top 10 hyperparameter configurations for the Asian dataset.

Rank	K	PCA	PCA Var.	Return Periods	Vol Windows	Silhouette	Sharpe	Combined
1	2	True	0.90	[1,5,21]	[21,63]	0.2029	4.3604	0.8282
2	2	True	0.95	[1,5,21]	[21,63]	0.1922	4.3670	0.8240
3	2	False	N/A	[1,5,21]	[21,63]	0.1845	4.3743	0.8213
4	2	True	0.90	[1,5,21,63]	[21,63]	0.2089	3.7526	0.7299
5	2	True	0.95	[1,5,21,63]	[21,63]	0.1997	3.7529	0.7254
6	2	False	N/A	[1,5,21,63]	[21,63]	0.1911	3.7575	0.7218
7	3	True	0.90	[1,5,21]	[21,63]	0.1689	3.7204	0.7045
8	3	True	0.95	[1,5,21]	[21,63]	0.1582	3.7316	0.7011
9	2	True	0.90	[1,5,21]	[10,21,63]	0.2173	3.5439	0.6993
10	2	True	0.95	[1,5,21]	[10,21,63]	0.2059	3.5613	0.6965

B. Team Work and Contributions

B.1. Author Contributions (CRediT)

Conceptualisation:

Methodology: Oscar Peyron

Implementation: Oscar Peyron

Data Processing & Feature Engineering: Oscar Peyron

Analysis: Oscar Peyron

Validation & Testing:

Visualisation:

Writing – Original Draft: Oscar Peyron

Writing – Review & Editing:

Project Administration:

Individual summary:

- **Zhidian Jing:** *[roles and concrete outputs to be added]*
- **Siqi Zhang:** *[roles and concrete outputs to be added]*
- **Stefaan Fernando:** *[roles and concrete outputs to be added]*
- **Daniel Greenstein:** *[roles and concrete outputs to be added]*
- **Oscar Peyron:** Implementation of the PCA K-Means pipeline for regime detection. This included data processing, feature engineering, model fitting, and backtesting of the trading strategy. Oscar performed preliminary hyperparameter tuning for the number of regimes. Wrote part of the original draft, particularly the methodology and results sections.

B.2. Reflection on Team Work

B.3. AI Statement

Generative AI tools (Claude, Gemini) were used in two capacities during this project:

1. **Code development:** AI assistance was used to generate baseline code, including boilerplate data-loading routines and initial model templates. All AI-generated code was reviewed, tested, and adapted by team members before inclusion in the final codebase.

2. **Report writing:** AI tools were used to assist with drafting and editing sections of this report, including improving clarity, grammar, and structure. All AI-generated text was critically reviewed and revised by the authors to ensure accuracy and consistency with the project's findings.